TrustGate IoT

Implementation and evaluation report | 31 October 2026

Team LinkX | A3-PS004-TC154 | CHRIST University

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Problem and implemented solution

PS004: Authenticated Gateway-Assisted DNSSEC for IoT Clients. A constrained device delegates signature-chain validation to a trusted gateway. The device must authenticate that gateway before relying on its DNSSEC decision.

The prototype uses a classic ESP32, Unbound 1.26.2 in Ubuntu/WSL, a Windows TCP forwarding rule and local BIND test zones. The ESP32 opens DNS over TLS on TCP 1853 and verifies the provisioned CA, hostname and certificate validity period. There is no plaintext fallback.

Data path and trust boundary

ESP32 -> authenticated TLS -> Windows forwarding -> Unbound -> loopback BIND test zones. Windows forwards TLS bytes; Unbound terminates TLS and validates DNSSEC. USB serial provisions the board clock after boot.

The ESP32 requires a successful DNS answer with the authenticated-data flag and a direct IPv4 address. It checks framing, response ID and question consistency. It does not independently validate DNSSEC signatures. A compromised trusted gateway is outside this design's protection.

Scope

The local zone uses a provisioned DNSSEC trust anchor. This is not a public-root validation chain. Test addresses are documentation fixtures, not reachable sensors. CNAME chains, IPv6 answers, secure boot, flash encryption and automated credential renewal are not implemented.

Verification evidence

The existing software suite covers 26 DNS, DNSSEC, TLS and resilience checks. Two additional network-configuration unit tests check safe address updates and preservation of credentials. Structured evidence is stored under results/.

Physical-board tests on 29 October accepted valid signed data, rejected unsigned and tampered data, and handled NXDOMAIN. Untrusted-issuer and wrong-hostname certificates were rejected with explicit verification errors. The genuine gateway was accepted again without resetting the board.

Two gateway restart cycles and two client-requested Wi-Fi reconnect cycles passed. These do not reproduce access-point power loss or long-duration field operation.

31 October run: 60 attempts over 166.78 seconds; 59 accepted and 1 blocked/failed. Startup attempts are included. Run complete: True.

Measurements and interpretation

Earlier ESP32 rapid-query phases accepted 48/48 measured requests after warm-up. Successful median elapsed time was 888 ms with modem sleep on and 805 ms with it off. Startup failures were excluded from that table and are disclosed in the measurement notes.

The isolated laptop/WSL cache experiment accepted 20/20 cold and 20/20 warm requests. Median times: 7.01 ms cold, 2.45 ms warm. Every request used a fresh TLS connection; BIND remained running.

These are different experiments and must not be compared as an offloading speedup. CPU savings, energy savings and a comparable client-side DNSSEC baseline have not been measured. Shared Wi-Fi, startup retries and an earlier incomplete response limit reliability claims.

Reproduce and review

Read docs/setup.md, run scripts/network-preflight.ps1, and compare the laptop, WSL and ESP32 addresses. scripts/configure-network.py updates local address settings without displaying Wi-Fi credentials. Apply the board-specific forwarding rule, rebuild/flash if the gateway address changed, then start the lab.

In Ubuntu: bash scripts/run-dnssec-stage.sh runs the local security suites. On Windows: python scripts/test-board-certificates.py tests physical-board certificate rejection; python scripts/test-board-recovery.py tests recovery; python scripts/soak-board.py records all repeated-query attempts.

For cold/warm cache measurements run .venv/bin/python scripts/benchmark-cache.py in Ubuntu. It starts temporary loopback-only resolvers and leaves the live gateway configuration unchanged.

Demonstration plan

Follow docs/demo-walkthrough.md: introduce the problem and schematic, identify the physical board, show labelled tiled terminals, demonstrate DNS policy and gateway authentication, then explain recovery, measurements and limitations. The existing YouTube introduction is not the final working demonstration.

Submission status and remaining work

Source and supporting files form an evaluator review package. A final working-demo recording or live URL still needs to be supplied by the team. Repository ownership and collaborator acceptance require current verification. Eligibility remains an organizer decision.

Further work: controlled-network tests, real access-point outage recovery, longer reliability runs, portable deployment and production provisioning. Do not claim that the intermittent failure cause is fully resolved.

References

Official guidelines: https://portal.aiori.in/aiori-3/hackathon-guideline/

RFC 4035: DNSSEC protocol modifications; RFC 7858: DNS over TLS; RFC 8310: authentication profiles.

Repository: https://github.com/Lalittt21/A3-PS004-TC154